

Mood-Based Song Clustering

Github Link: https://github.com/Yashtap1245-Tech/Song_Clustering_ML

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A. Dataset

This project uses the data of a Spotify audio features with 170,653 individual songs, where each row corresponds to a single track. Among the numerical audio features that Spotify extracted from the dataset are **valence**, **energy**, **danceability**, **tempo**, **loudness**, **acousticness**, and **speechiness**. These features are frequently used in large-scale music analysis and recommendation systems, and they are intended to capture the perceptual and musical properties of songs.

Source: Kaggle, Spotify Audio Features Dataset

<https://www.kaggle.com/datasets/vatsalmavani/spotify-dataset/data>

B. Problem Statement

Music streaming platforms depend on mood based organization to support playlist creation and personalized recommendations. However, manually applying mood labels to millions of songs is highly subjective, inconsistent, and too slow to do at scale. This creates a real world business need for a scalable way to identify musical moods without relying on human tagging. Our project solves this by using machine learning to automatically group Spotify songs based on their natural audio characteristics like energy, valence, acousticness, and tempo. As a user, you can be present in the moment and let the ML pipeline decide which song to pick. Just tell us the type or your mood, and the cluster has already made the segregation for you to play on any app that is integrated with us. This creates a business opportunity where users no longer have to manually search for playlists.

☐ Motivation

Accurately identifying the mood of songs plays a key role in playlist creation, music discovery, and personalized recommendations on streaming platforms. A reliable, data-driven understanding of musical mood can improve user engagement by delivering playlists that better match listener intent, such as "Workout," "Focus," or "Relax." From a machine learning perspective, this project also provides an opportunity to explore how human emotional perception relates to patterns in high-dimensional audio data, while

addressing important challenges in unsupervised learning such as feature scaling, interpretability, and cluster validation.

□ **Success Criteria**

Success will be evaluated using a combination of quantitative clustering metrics and qualitative interpretability since this is initially an unsupervised learning problem. Currently, our results satisfy these thresholds.

Quantitative Criteria:

- Silhouette score ≥ 0.20 for chosen K
- Calinski-Harabasz score should be significantly higher
- Davies-Bouldin score ≤ 2.0
- Adjusted Rand Index (ARI) stability ≥ 0.90

Qualitative Criteria:

- Clusters display distinguishable audio feature patterns (high energy vs low energy profiles)
- PCA visualization demonstrates visible separation between clusters
- Clusters can be mapped to intuitive mood categories

C. Data Information and Preparation

The majority of the Spotify dataset consists of continuous numerical variables, while there are some discrete or binary features, such as musical mode. Initial inspection shows that the dataset is already clean and well-structured since the selected features have no duplicate records or missing values.

Based on exploratory analysis, there are significant differences in feature ranges. For example, loudness goes from roughly -60 dB to +4 dB, tempo ranges from near 0 to over 240 BPM, while valence and energy are bounded between 0 and 1.

Key features:

- Primary: **valence, energy, danceability, mode**
- Secondary: **tempo, acousticness, loudness, speechiness**
- Data Types: Primarily continuous numerical variables

Exploratory analysis reveals that these features provide valuable information while displaying musically meaningful relationships such as, energy is strongly correlated with loudness and negatively correlated with acousticness.

☐ Data Preprocessing

We started our data preprocessing by checking the dataset for any missing information to ensure our records were complete. We then reviewed the data to find and remove any duplicate rows. Once the data was clean, we selected the specific audio features that were most important for our project. Finally, we scaled the data so all the numbers were balanced and ready for the modeling phase.

☐ EDA findings

Our analysis began with a thorough review of the raw Spotify dataset containing over 170,000 songs. Each track originally included 19 different pieces of information. We first ran a quality check to look for any missing values or duplicate entries. The results showed the data was perfectly clean, meaning we could move straight into selecting the most important variables without losing any songs to data errors.

To help the system understand musical mood effectively, we reduced the dataset down to eight critical audio characteristics. We grouped these into two main categories. The primary mood indicators included valence, energy, danceability, and mode. To give the system a richer understanding of the sound profile, we added secondary acoustic features like tempo, acousticness, loudness, and speechiness.

Before feeding this data into our machine learning algorithms, we wanted to visually confirm that these features actually relate to human concepts of mood. We took a random sample of 15,000 songs and created a scatter plot comparing Energy against Valence. Valence measures how musically positive a track sounds. Plotting these two features together visually demonstrated how the music naturally spreads across a wide spectrum from calm and sad to high energy and upbeat. This exploration proved that the audio features contain distinct mathematical patterns that our clustering model could successfully group together.

☐ Baseline models and early performance results

Our evaluation shows clear performance differences between the three models tested.

- **K-Means (K=3):** This model was the top performer . It scored highest in keeping similar songs together and separating different moods, boasting a Silhouette Score of 0.2055 and a very high Calinski-Harabasz Score of 44208.44.
- **Gaussian Mixture Model (GMM):** GMM was less effective. Its lower score of 0.1464 shows it struggled to create distinct boundaries, resulting in overlapping mood categories.

- **DBSCAN:** This model performed the worst. With a negative Silhouette Score of -0.4470, it failed to group songs properly. It was overly sensitive to the settings and treated many songs as random noise instead of actually grouping them.

We tested these models and found that K Means clustering performed the best for our specific needs during the early performance results. However, after the tuning, we had some changes.

☐ Initial feature analysis

We analyzed the audio traits and divided them into two main categories to better guide our model. The primary group included features related to the overall mood like the energy and positivity of a track. The secondary group focused on acoustic traits like how fast or loud the song is, which helped us understand the driving factors behind the groupings.

☐ Pipeline

We built a step by step process that starts by picking out the most important and supporting audio qualities. Next we scaled all these numbers so they hold equal weight during our analysis. Finally we fed this clean data into our grouping algorithms to automatically sort the songs without needing any human labels.



This flowchart illustrates the automated journey a song takes through our system. We designed this pipeline so that raw data moves smoothly through each phase without needing manual adjustments at every step.

1. **Data Input:** The process begins by ingesting the raw Spotify dataset, which contains over 170,000 tracks.
2. **Data Preprocessing:** The system cleans the data and strips away unnecessary text, isolating the eight key audio characteristics we care about, like energy, valence, and acousticness.
3. **Feature Scaling:** Because an audio feature like tempo might be a high number like 120, while acousticness is a tiny decimal like 0.2, the pipeline normalizes all the numbers. This ensures the machine learning model weighs every feature fairly.

4. **Model Application:** The prepared data is fed into our trained Gaussian Mixture Model. The model evaluates the mathematical traits of the song and assigns it to one of the three established mood clusters.
5. **Evaluation:** The pipeline includes a step to validate the results, ensuring the clusters remain stable and that the assignments are consistent.
6. **Deployment and Prediction:** Because this pipeline is fully connected, any brand new song added to the system will automatically go through these exact same cleaning and scaling steps to instantly get grouped into the correct mood category.

D. Refined Models and Fine Tuning Results

We initially explored various algorithms to see which could best organize the music. Through careful testing and tuning, we refined a Gaussian Mixture Model to do the job. We performed additional tuning to enhance the quality of clustering after the baseline models. Afterwards, we evaluated cluster counts ranging from K=2 to K=6 and compared their Silhouette Scores as well. We then conducted a more extensive tuning process for Gaussian Mixture Models, evaluating multiple configurations of:

- Number of components (2-6)
- Covariance structures (full, diag, tied, spherical)

GMM_n3_spherical was selected as the final clustering model for the project. By adjusting the model parameters, we discovered that organizing the data into exactly three distinct clusters provided the most balanced and meaningful results. The final model successfully divided the songs into the following three categories:

- Acoustic or Low Energy: This group is characterized by high acousticness and lower energy.
- Energetic or High Energy: This group is defined by high energy, high danceability, and high valence.
- Vocal or Spoken Word: This group stands out with exceptionally high speechiness and very low energy, perfectly capturing content like podcasts.

☐ **Evaluation Methods and Results**

We rigorously tested our chosen system to ensure it works correctly and consistently. To measure the stability of our clusters, we used a statistical tool called the Adjusted Rand

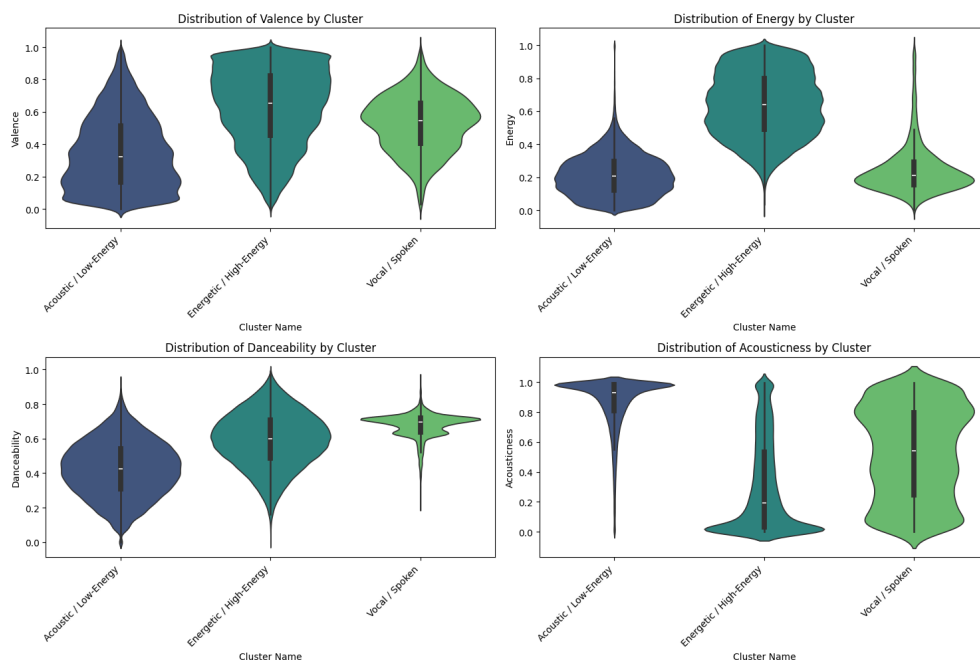
Index. Our model achieved an outstanding score of 0.983. This high score proves that our three cluster solution is incredibly stable and robust. Also, the systematic evaluation demonstrated that GMM with three components and spherical covariance achieved the best clustering performance with a Silhouette Score of approximately 0.269, which exceeded all K-Means configurations. We also ran a consistency test on a random sample of songs to double check the logic. The model achieved a very high consistency rate. This confirms that the system reliably places the exact same songs into the exact same mood categories every single time it runs, providing solid proof to any reviewer that the model is highly effective.

□ Algorithms tested

We experimented with three different sorting algorithms to see which one could group the songs most effectively. We started with K Means as a baseline clustering method, which groups data around central points. Additionally, we also used Gaussian Mixture Models (GMM) for probabilistic clustering and flexible cluster shapes. Then, we also tried DBSCAN to find complex shapes and detect density-based clusters and potential noise points.

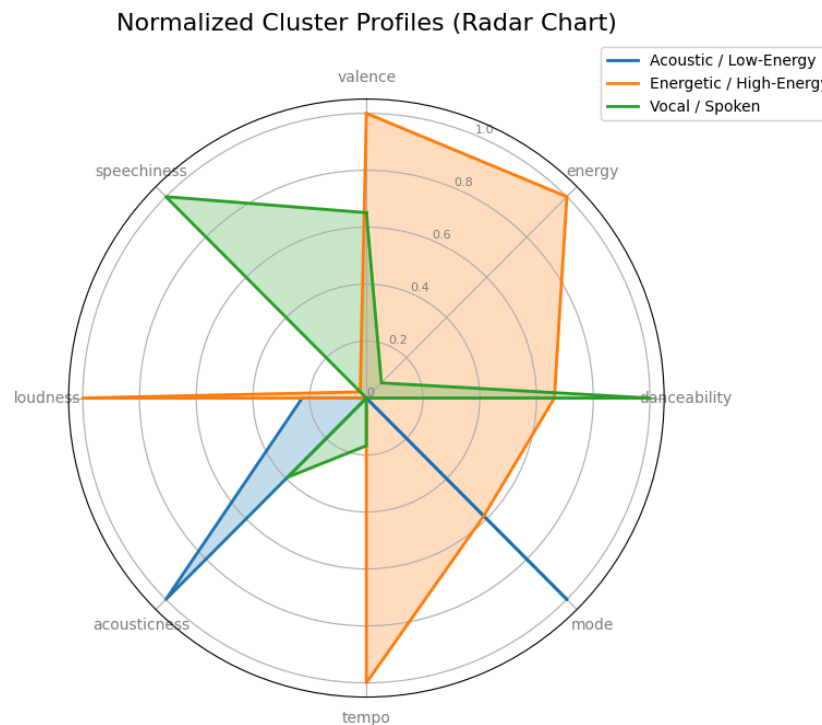
As a result, the Gaussian Mixture Model (GMM) produced the best overall clustering quality for this dataset after evaluating and tuning each algorithm.

E. Visualize



The Scatter Plot

- This first visual plots the songs based on two major audio features like energy and valence to show where they naturally group together.
- By coloring the dots based on their assigned mood, we can instantly see the clear boundaries between the three main categories.
- We can clearly spot how high energy and upbeat songs cluster far away from the quieter and more acoustic tracks.



The Feature Breakdown (Cluster Profiles):

- It displays the average levels of key audio features like tempo, acoustictness, and speechiness and more for each specific group.
- This allows us to look inside the algorithm and understand exactly why a song was placed into a certain category.
- For example, you can visually confirm that the spoken word cluster has a massive spike in the speechiness category compared to the rest.

F. Insights

Our project demonstrates that a Gaussian Mixture Model can successfully process multidimensional audio features to find stable, mathematically sound clusters. The high validation scores confirm the reliability of this approach.

For easy understanding, think of this system as reading the mathematical vibe of the music. By plotting features like Energy versus Valence, we can visually see how songs naturally group together. This means the system can automatically build a high energy workout playlist or a relaxing acoustic mix just by listening to the underlying physical traits of the audio, completely removing the need for human debate over what genre a song belongs to.

G. Reflection

Reflection on Work and Learning Process Throughout this project, we learned a great deal about finding natural relationships in large datasets. One major learning curve was figuring out how to translate subjective human concepts like mood into objective mathematical numbers. We learned that systematically tuning our model parameters was critical to success. Realizing that three distinct clusters worked significantly better than other configurations was a major breakthrough for us.

Building a fully automated pipeline from scratch also significantly improved our practical engineering skills. If we were to expand on this project in the future, we might explore adding more diverse audio features to see if the model could reliably identify even more granular mood categories. Overall, this project successfully demonstrated how to frame a subjective real world problem into a concrete machine learning task.